

Draft Standard for Local and metropolitan area networks —

Frame Replication and Elimination for Reliability

Corrigendum 1: Technical and Editorial Corrections

Prepared by the Maintenance Task Group of IEEE 802.1

LAN **MAN** **Standards** **Committee**
of
IEEE Computer Society the

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Abstract: This corrigendum to IEEE Std 802.1CB™-2017 provides technical and editorial corrections.

Keywords: TSN, Time-Sensitive Networking, Redundancy, Bridging, Bridges, Frame Replication, Frame Elimination, Stream identification, Extended Stream identification, Bridged Local Area Networks, IEEE 802®, IEEE 802.1Q™, IEEE 802.1CB™, IEEE 802.1CBcv™, IEEE 802.1CBdb™, local area networks (LANs), MAC Bridges, Virtual Bridged Local Area Networks (virtual LANs), amendment.

1 Editor's Foreword

2 <<Notes>>

3 <<Throughout this document, all notes such as this one, presented between angle braces, are temporary
4 notes inserted by the Editors for a variety of purposes; these notes and the Editors' Foreword will all be
5 removed prior to publication and are not part of the normative text.>>

6 <<Comments and participation in 802.1 standards development

7 Comments on this draft are encouraged. **PLEASE NOTE: All issues related to IEEE standards
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11 only to comments that materially affect either the technical content of the document or the clarity of that
12 technical content. Comments should not simply state what is wrong, but also what might be done to fix the
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16 website:

17 <https://1.ieee802.org/>

18 Use of the email distribution list is not presently restricted to 802.1 members, and the working group has had
19 a policy of considering ballot comments from all who are interested and willing to contribute to the
20 development of the draft. Individuals not attending meetings have helped to identify sources of
21 misunderstanding and ambiguity in past projects. Non-members are advised that the email lists exist primarily
22 to allow the members of the working group to develop standards, and are not a general forum.

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24 familiarize themselves with that policy. See <http://standards.ieee.org/about/sasb/patcom/materials.html>.

25 Comments on this document may be sent to the 802.1 email exploder, to the editors, or to the Chairs of the
26 802.1 Working Group and Time-Sensitive Networking Task Group.

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33 **PLEASE NOTE: Comments whose distribution is restricted in any way cannot be considered, and
34 may not be acknowledged. >>**

35 <<The draft text and accompanying information

36 This document currently comprises:

- 37 — A title page for the proposed standard including an Abstract and Keywords. This title page will be retained following working group approval of this draft, i.e. prior to sponsor ballot.

- 1 — The editors' forewords, including this text. These include an unofficial and informal appraisal of history and status, introductory notes to each draft that summarize the progress and focus of each successive draft, and requests for comments and contributions on major issues.
- 4 — IEEE boilerplate text.
- 5 — A record of participants (not included in early drafts but added prior to publication).
- 6 — The introduction to this standard.
- 7 — The proposed standard proper.

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9 coherent drafts from the resolutions of ballot comments and the other discussions that take place in the
10 working group meetings. Preparation of drafts often exposes inconsistencies in editors instructions or
11 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
12 and requests by the editors' for contributions on specific issues will be found in the editors' introductory notes
13 to the current draft, at appropriate points in the draft, and in Annex Z. Significant discussion of more difficult
14 topics will be found in the last of these.

15 The ballot comments received on each draft, and the editors' proposed and final disposition of comments, are
16 part of the audit trail of the development of the standard and are available, along with all the revisions of the
17 draft on the 802.1 web site (for address see above).

18 >>

19 << **Introductory notes to draft 1.1**

20 Draft 1.1 was prepared for a first Working Group recirculation ballot, as a result of comment resolution on
21 Draft 1.0, carried out at the IEEE 802.1 May 2025 interim meeting. The ballot comment resolution is
22 documented in 802-1CB-2017-cor1-d1-0-dis-v01.pdf. Changes from Draft 1.0 include:

- 23 — typo and text consistency corrections;
- 24 — As maintenance item #343 is out of the PAR's scope, all clauses impacted by the changes introduced
25 by its resolution are removed from this draft, i.e. clauses 6.3, 6.4, 6.5, 6.8, 8 and Annex C. The
26 resolutions of the comments made against these clauses are kept for implementation in another
27 vehicle document.

28 >>

29 << **Introductory notes to draft 1.0**

30 Draft 1.0 was prepared for a first Working Group ballot, as a result of comment resolution on Draft 0.1, carried
31 out at the IEEE 802.1 March 2025 virtual meeting. The ballot comment resolution is documented in 802-1CB-
32 2017-cor1-d0-1-dis-v01.pdf. Changes from Draft 0.1 include:

- 33 — various typo corrections;
- 34 — Consistency between modified figures and corresponding descriptive text;
- 35 — Alignment of the Stream_handle value definition between clauses 6.2 and 9.1.1.1.

36 >>

37 << **Project Authorization Request, Scope, Purpose, and Five Criteria**

38 A PAR (Project Authorization Request) for P802.1CB-2017/Cor1 was approved by the IEEE Standards
39 Association on February 15, 2024. The following information is taken from the 802.1CB-2017/Cor1 PAR.

1 Scope of Proposed Changes:

2 Correction of technical and editorial errors identified and agreed by the IEEE 802.1 Working Group
3 maintenance activity comprising: <https://www.802-1.org/items/431>, <https://www.802-1.org/items/451>, [https://](https://www.802-1.org/items/456)
4 www.802-1.org/items/456, and [https://www.802-1.org/ items/471](https://www.802-1.org/items/471). The management module changes needed
5 are applicable in the context of IEEE Std 802.1CBv(TM)-2021..

6 Purpose of Proposed Project:

7 This document will not include a purpose clause.

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42

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1 Participants

2 The following is a list of participants in the IEEE 802.1 Working Group.

3 **Glenn Parsons, *Chair***
4 **Jessy Rouyer, *Vice-Chair***
5 **Mark Hantel, *Chair, Maintenance Task Group***
6 **Christophe Mangin, *Editor***
7

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SBMember8	SBMember16	

2 *Member Emeritus
3

4 Also included are the following nonvoting IEEE-SA Standards Board liaisons:

5
6 Name, *DOE Representative*
7 Name, *NIST Representative*
8
9
10 Name
11 IEEE Standards Program Manager, Document Development
12
13
14 Name
15 IEEE Standards Program Manager, Technical Program Development

1 Introduction

2

This introduction is not part of IEEE P 802.1CB-2017/Cor1-20XX, IEEE Draft Standard for Local and metropolitan area networks – Frame Replication and Elimination for Reliability – Corrigendum 1.

3 The first edition of IEEE Std 802.1CB was published in 2017. A first amendment IEEE Std 802.1CBv-
4 2021 added the YANG data model that allows configuration and state reporting for all managed objects of
5 the base standard. A second amendment IEEE Std 802.1CBdb-2021 added the mask-and-match Stream
6 identification function and provided technical and editorial corrections.

7 This corrigendum IEEE 802.1CB-2017/Cor1-2025 implements the changes necessitated by various mainte-
8 nance items that have been notified to the IEEE 802.1 Working Group. The maintenance items addressed by
9 this corrigendum are: 315, 338, 344 and 364.

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1

2 Draft Standard for 3 Local and metropolitan area networks — 4 Frame Replication and Elimination for 5 Reliability 6 Corrigendum 1: Technical end Editorial 7 Corrections

8 NOTE—The editing instructions contained in this amendment define how to merge the material contained here into the
9 base document and its other amendments to form the new comprehensive standard.

10 The editing instructions are shown in ***bold italic***. Four editing instructions are used: ***change***, ***delete***, ***insert***, and ***replace***.
11 ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change
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17 incorporated into the base standard.⁶

⁶Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

1 6. Stream identification

2 6.2 Stream identification function

3 *Insert the following paragraph at the end of the clause, before the notes, in IEEE Std*
4 *802.1CB-2017:*

5 The null-value is a value known from the management system as identifying the stream_handle assigned to
6 packets that belong to no known Stream.

7 6.7 IP Stream identification

8 *Insert the following text after the second paragraph:*

9 Practically, the IP Stream identification function requires locating the beginning of the IP packet
10 encapsulated in the frame, preceded by an IP EtherType field, e.g., 0x0800 or 0x88DD. The location of the
11 IP EtherType field within the frame is dependent on the combination of tag(s) following the addresses in a
12 frame. The method used to determine this location is beyond the scope of this standard.

13

1 9. Stream Identification Management

2 9.1 Stream identity table

3 9.1.1 tsnStreamIdEntry

4 9.1.1.1 tsnStreamIdHandle

5 *Insert the following note at the end of the subclause:*

6 NOTE—It is also possible to assign a packet that was identified as belonging to a known stream (i.e., a specific tsn-
7 StreamIdEntry) a stream_handle equal 0.

8

1 12. YANG Data Model

2 12.1 YANG Framework

3 *Change the text in the lettered-list as shown:*

4 This clause has been developed according to the YANG guidelines published in IETF RFC 6087 [B5] as
5 applicable to IEEE standards.

6 The YANG framework applies hierarchy in the following areas:

- 7 a) The uniform resource name (URN), as specified in IEEE Std 802d [B2a]. ~~The structure of the URN~~
8 ~~is such that ieee is the root (i.e., name-space identifier), followed by the standard, then the working~~
9 ~~group developing the standard.~~
- 10 b) The YANG objects form a hierarchy of configuration and operational data structures that define the
11 YANG model. These hierarchical relationships are described in 12.2.

12 12.2 IEEE Std 802.1CB YANG model

13 12.2.2 Frame Replication and Elimination for Reliability model

14 *Replace Figure 12-5 with the following figure:*

1

sequence-generation (10.3)		
uint32	index	// r-w
handle-ref	* stream	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w

sequence-recovery (10.4)		
uint32	index	// r-w
handle-ref	* stream	// r-w
if-ref	* port	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
uint32	reset-timeout	// r-w
uint32	invalid-sequence-value	// r
boolean	take-no-sequence	// r-w
boolean	individual-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint16	latent-error-paths	// r-w
uint32	latent-error-reset-period	// r-w

sequence-identification (10.5)		
handle-ref	* stream	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
boolean	active	// r-w
enum	encapsulation	// r-w
int8	path-id-lan-id	// r-w

stream-split (10.6)		
uint32	index	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
leaf-ref	* input-id	// r-w
leaf-ref	* output-id	// r-w

sequence-autoconfiguration (10.7.1)		
uint32	index	// r-w
enum	sequence-encapsulation	// r-w
if-ref	* receive-port	// r-w
enum	tagged	// r-w
dot1qtypes:vlanid	* vlan	// r-w
if-ref	* recovery-port	// r-w
uint64	destruction-interval	// r-w
uint64	reset-interval	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
boolean	create-individual	// r-w
boolean	create-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint32	latent-error-reset-period	// r-w

output-autoconfiguration (10.7.2)		
uint32	index	// r-w
if-ref	* port	// r-w
enum	encapsulation	// r-w
int8	lan-path-id	// r-w

Figure 12-5—FRER model

1 12.6 Definition of 802.1CB YANG modules^{7 8}**2 *Change clause header as shown below:*****3 12.6 Definition of [IEEE Std](#) 802.1CB YANG modules****4 12.6.1.2 YANG data scheme definition for ieee802-dot1cb-frer YANG module****5 *Change the text as shown:***

```

6 module: ieee802-dot1cb-frer
7   +--rw frer
8     +--rw sequence-generation* [index]
9   | +--rw index                               uint32
10  |   +--rw stream*                           -> /dot1cb-sid:stream-identity/handle
11  |   +--rw direction-out-facing?             dot1cb-sid-types:direction
12  |   +--rw reset?                            boolean
13  +--rw sequence-recovery* [index]
14  |   +--rw index                               uint32
15  |   +--rw stream*
16  |   |   -> /dot1cb-sid:stream-identity/handle
17  |   +--rw port*                             if:interface-ref
18  |   +--rw direction-out-facing?             dot1cb-sid-types:direction
19  |   +--rw reset?                            boolean
20  |   +--rw algorithm
21  |   |   +--rw (algorithm)?
22  |   |   |   +--:(vector)
23  |   |   |   |   +--rw vector!
24  |   |   |   |   |   +--ro type-number?    dot1cb-frer-types:seq-rcvy-algorithm
25  |   |   |   |   |   +--ro oui-cid?       string
26  |   |   |   |   +--:(match)
27  |   |   |   |   |   +--rw match!
28  |   |   |   |   |   +--ro type-number?    dot1cb-frer-types:seq-rcvy-algorithm
29  |   |   |   |   |   +--ro oui-cid?       string
30  |   |   |   |   +--:(organization-specific)
31  |   |   |   |   |   +--rw organization-specific!
32  |   |   |   |   |   +--rw type-number?    int32
33  |   |   |   |   |   +--rw oui-cid?       string
34  |   +--rw history-length?                   sequence-history-length
35  |   +--rw reset-timeout?                   uint32
36  |   +--ro invalid-sequence-value?          uint32
37  |   +--rw take-no-sequence?                boolean
38  |   +--rw individual-recovery?             boolean
39  |   +--rw latent-error-detection?          boolean
40  |   +--rw latent-error-detection-parameters
41  |   |   +--rw difference?                   int32
42  |   |   +--rw period?                      uint32
43  |   |   +--rw paths?                      uint16
44  |   |   +--rw reset-period?                uint32
45  +--rw sequence-identification* [port direction-out-facing]
46  |   +--rw stream*                           -> /dot1cb-sid:stream-identity/handle
47  |   +--rw port                             if:interface-ref

```

⁷Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁸An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

1 | +--rw direction-out-facing      dot1cb-sid-types:direction
2 | +--rw active?                  boolean
3 | +--rw encapsulation
4 | | +--rw (encapsulation)?
5 | | | +--:(r-tag)
6 | | | | +--rw r-tag!
7 | | | | +--ro type-number?      dot1cb-frer-types:seq-encaps-method
8 | | | | +--ro oui-cid?          string
9 | | | +--:(hsr-sequence-tag)
10 | | | | +--rw hsr-sequence-tag!
11 | | | | +--ro type-number?      dot1cb-frer-types:seq-encaps-method
12 | | | | +--ro oui-cid?          string
13 | | | +--:(prp-sequence-tag)
14 | | | | +--rw prp-sequence-tag!
15 | | | | +--ro type-number?      dot1cb-frer-types:seq-encaps-method
16 | | | | +--ro oui-cid?          string
17 | | | +--:(organization-specific)
18 | | | | +--rw organization-specific!
19 | | | | +--rw type-number?      int32
20 | | | | +--rw oui-cid?          string
21 | +--rw path-id-lan-id?          lan-path-id
22 +--rw stream-split* [port direction-out-facing index]
23 | +--rw index                  uint32
24 | +--rw port                    if:interface-ref
25 | +--rw direction-out-facing    dot1cb-sid-types:direction
26 | +--rw input-id*               -> /dot1cb-sid:stream-identity/handle
27 | +--rw output-id*              -> /dot1cb-sid:stream-identity/handle
28 +--rw autoconfiguration {auto-configuration}?
29 | +--rw sequence* [index]
30 | | +--rw index                  uint32
31 | | +--rw sequence-encapsulation
32 | | | +--rw (encapsulation)?
33 | | | | +--:(r-tag)
34 | | | | | +--rw r-tag!
35 | | | | | +--ro type-number?
36 | | | | | | dot1cb-frer-types:seq-encaps-method
37 | | | | | +--ro oui-cid?          string
38 | | | | +--:(hsr-sequence-tag)
39 | | | | | +--rw hsr-sequence-tag!
40 | | | | | +--ro type-number?
41 | | | | | | dot1cb-frer-types:seq-encaps-method
42 | | | | | +--ro oui-cid?          string
43 | | | | +--:(prp-sequence-tag)
44 | | | | | +--rw prp-sequence-tag!
45 | | | | | +--ro type-number?
46 | | | | | | dot1cb-frer-types:seq-encaps-method
47 | | | | | +--ro oui-cid?          string
48 | | | | +--:(organization-specific)
49 | | | | | +--rw organization-specific!
50 | | | | | +--rw type-number?      int32
51 | | | | | +--rw oui-cid?          string
52 | +--rw receive-port*           if:interface-ref
53 | +--rw tagged?                 enumeration
54 | +--rw vlan*                   dot1qtypes:vlanid
55 | +--rw recovery-port*          if:interface-ref
56 | +--rw destruction-interval?   uint64
57 | +--rw reset-interval?         uint64
58 | +--rw algorithm
59 | | +--rw (algorithm)?

```

```

1      | |      +---: (vector)
2      | |      | +---rw vector!
3      | |      | +---ro type-number?
4      | |      | |      dotlcb-frer-types:seq-rcvy-algorithm
5      | |      | +---ro oui-cid?      string
6      | |      +---: (match)
7      | |      | +---rw match!
8      | |      | +---ro type-number?
9      | |      | |      dotlcb-frer-types:seq-rcvy-algorithm
10     | |      | +---ro oui-cid?      string
11     | |      +---: (organization-specific)
12     | |      +---rw organization-specific!
13     | |      +---rw type-number?    int32
14     | |      +---rw oui-cid?      string
15     | +---rw history-length?      sequence-history-length
16     | +---rw create-individual?    boolean
17     | +---rw create-recovery?      boolean
18     | +---rw latent-error-detection? boolean
19     | +---rw latent-error-difference? int32
20     | +---rw latent-error-period?   uint32
21     | +---rw latent-error-reset-period? uint32
22     +---rw output* [index]
23         +---rw index                uint32
24         +---rw port*                if:interface-ref
25         +---rw encapsulation
26         | +---rw (encapsulation)?
27         | +---: (r-tag)
28         | | +---rw r-tag!
29         | | +---ro type-number?
30         | | |      dotlcb-frer-types:seq-encaps-method
31         | | +---ro oui-cid?      string
32         | +---: (hsr-sequence-tag)
33         | | +---rw hsr-sequence-tag!
34         | | +---ro type-number?
35         | | |      dotlcb-frer-types:seq-encaps-method
36         | | +---ro oui-cid?      string
37         | +---: (prp-sequence-tag)
38         | | +---rw prp-sequence-tag!
39         | | +---ro type-number?
40         | | |      dotlcb-frer-types:seq-encaps-method
41         | | +---ro oui-cid?      string
42         | +---: (organization-specific)
43         | +---rw organization-specific!
44         | +---rw type-number?    int32
45         | +---rw oui-cid?      string
46         +---rw lan-path-id?      lan-path-id
47
48     augment /dotlcb-sid:stream-identity:
49         +---ro auto-configured?    boolean {auto-configuration}?
50         +---rw lan-path-id?        lan-path-id {auto-configuration}?
51     augment /if:interfaces/if:interface/if:statistics:
52         +---ro frer
53         +---ro per-port-counters
54         | +---ro rx-passed-pkts?    uint64
55         | +---ro rx-discarded-pkts? uint64
56         | +---ro encode-errored-pkts? uint64
57         +---ro per-port-per-stream-counters* [direction-out-facing handle]
58         +---ro direction-out-facing    dotlcb-sid-types:direction
59         +---ro handle

```

```

1      |      -> /dot1cb-sid:stream-identity/handle
2      +--ro generation-reset?          uint64
3      +--ro rx-out-of-order-pkts?      uint64
4      +--ro rx-rogue-pkts?             uint64
5      +--ro rx-passed-pkts?            uint64
6      +--ro rx-discarded-pkts?         uint64
7      +--ro rx-lost-pkts?              uint64
8      +--ro rx-tagless-pkts?           uint64
9      +--ro rx-resets?                  uint64
10     +--ro rx-latent-error-resets?     uint64
11     +--ro encode-errored-pkts?        uint64

```

12

13 12.6.2 YANG data module definitions

14 12.6.2.4 Definition for the ieee802-dot1cb-frer YANG module

15 *Change the text as shown:*

```

16 module ieee802-dot1cb-frer {
17   yang-version "1.1";
18   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1cb-frer;
19   prefix dot1cb-frer;
20   import ieee802-dot1cb-stream-identification {
21     prefix dot1cb-sid;
22   }
23   import ieee802-dot1cb-stream-identification-types {
24     prefix dot1cb-sid-types;
25   }
26   import ieee802-dot1cb-frer-types {
27     prefix dot1cb-frer-types;
28   }
29   import ieee802-dot1q-types {
30     prefix dot1qtypes;
31   }
32   import ietf-interfaces {
33     prefix if;
34   }
35   organization
36     "Institute of Electrical and Electronics Engineers";
37   contact
38     "WG-URL: http://ieee802.org/1/
39     WG-EMail: stds-802-1-1@ieee.org
40
41     Contact: IEEE 802.1 Working Group Chair
42     Postal: C/O IEEE 802.1 Working Group
43             IEEE Standards Association
44             445 Hoes Lane
45             Piscataway, NJ 08854
46             USA
47
48     E-mail: stds-802-1-chairs@ieee.org";
49   description
50     "Management objects that control the frame replication and
51     elimination from IEEE Std 802.1CB-2017. This YANG data model conforms
52     to the Network Management Datastore Architecture defined in RFC 8342.
53     Copyright (C) IEEE (2021). This version of this YANG module is part

```

```
1 of IEEE Std 802.1CBcv; see the standard itself for full legal
2 notices.";
3 revision 2024-08-22 {
4   description
5     "Published as part of IEEE Std 802.1CB-2017/Cor 1-2025.";
6   reference
7     "IEEE Std 802.1CB-2017/Cor 1-2025, Frame Replication and Elimination for
8     Reliability - Corrigendum 1.";
9 }
10 revision 2021-12-08 {
11   description
12     "Published as part of IEEE Std 802.1CBcv-2021. Initial version.";
13   reference
14     "IEEE Std 802.1CBcv-2021, Frame Replication and Elimination for
15     Reliability - FRER YANG Data Model and Management Information Base
16     Module.";
17 }
18 feature auto-configuration {
19   description
20     "Autoconfiguration of entries in the tables for Stream identity
21     table, sequence recovery table, and sequence identification table.";
22   reference
23     "7.11 of IEEE Std 802.1CB-2017
24     10.7 of IEEE Std 802.1CB-2017";
25 }
26 typedef lan-path-id {
27   type int8;
28   description
29     "An integer specifying a path or LAN. If and only if a packet
30     matches an entry in the Sequence identification table that
31     specifies HSR or PRP in its frerSeqEncEncapsType object,
32     tsnStreamIdLanPathId specifies the LanId or PathId value that must
33     be matched for this tsnStreamIdEntry to apply. A value of -1
34     indicates that the LanId or PathId are to be ignored.";
35   reference
36     "10.22 of IEEE Std 802.1CB-2017";
37 }
38 typedef sequence-history-length {
39   type uint32 {
40     range "2..max";
41   }
42   description
43     "An integer specifying how many bits of the SequenceHistory
44     variable are to be used.";
45   reference
46     "7.4.3.2.2 of IEEE Std 802.1CB-2017";
47 }
48 grouping sequence-encode-decode {
49   description
50     "The sequence-encode-decode grouping indicates the type of
51     encapsulation used for this instance of the Sequence encode/decode
52     function.";
53   reference
54     "10.5.1.5 of IEEE Std 802.1CB-2017";
55   choice encapsulation {
56     description
57       "This choice indicates the type of encapsulation used for this
58       instance of the Sequence encode/decode function. The
59       encapsulation includes an Organizationally Unique Identifier
```

```
1      (OUI) or Company ID (CID) to identify the organization defining
2      the Sequence encode/decode method.";
3      reference
4      "10.5.1.5 of IEEE Std 802.1CB-2017";
5      container r-tag {
6          presence "true";
7          description
8              "R-TAG";
9          reference
10             "7.8 of IEEE Std 802.1CB-2017";
11          leaf type-number {
12              type dot1cb-frer-types:seq-encaps-method;
13              default "r-tag";
14              config false;
15              description
16                  "The type number used for the R-TAG encode/decode method.";
17              reference
18                  "10.5.1.5 of IEEE Std 802.1CB-2017";
19          }
20          leaf oui-cid {
21              type string {
22                  pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
23              }
24              default "00-80-C2";
25              config false;
26              description
27                  "The Organizationally Unique Identifier (OUI) or Company ID
28                  (CID) to identify the organization defining the encode/decode
29                  method. For encode/decode methods defined in IEEE Std
30                  802.1CB-2017 the OUI/CID is always 00-80-C2.";
31              reference
32                  "10.5.1.5 of IEEE Std 802.1CB-2017";
33          }
34      }
35      container hsr-sequence-tag {
36          presence "true";
37          description
38              "HSR sequence tag";
39          reference
40              "7.9 of IEEE Std 802.1CB-2017";
41          leaf type-number {
42              type dot1cb-frer-types:seq-encaps-method;
43              default "hsr-seq-tag";
44              config false;
45              description
46                  "The type number used for the HSR sequence tag encode/decode
47                  method.";
48              reference
49                  "10.5.1.5 of IEEE Std 802.1CB-2017";
50          }
51          leaf oui-cid {
52              type string {
53                  pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
54              }
55              default "00-80-C2";
56              config false;
57              description
58                  "The Organizationally Unique Identifier (OUI) or Company ID
59                  (CID) to identify the organization defining the encode/decode
```

```
1         method. For encode/decode methods defined in IEEE Std
2         802.1CB-2017 the OUI/CID is always 00-80-C2.";
3     reference
4         "10.5.1.5 of IEEE Std 802.1CB-2017";
5 }
6 }
7 container prp-sequence-tag {
8     presence "true";
9     description
10        "PRP sequence trailer";
11     reference
12        "7.10 of IEEE Std 802.1CB-2017";
13     leaf type-number {
14         type dot1cb-frer-types:seq-encaps-method;
15         default "prp-seq-trailer";
16         config false;
17         description
18            "The type number used for the PRP sequence trailer
19            encode/decode method.";
20         reference
21            "10.5.1.5 of IEEE Std 802.1CB-2017";
22     }
23     leaf oui-cid {
24         type string {
25             pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
26         }
27         default "00-80-C2";
28         config false;
29         description
30            "The Organizationally Unique Identifier (OUI) or Company ID
31            (CID) to identify the organization defining the encode/decode
32            method. For encode/decode methods defined in IEEE Std
33            802.1CB-2017 the OUI/CID is always 00-80-C2.";
34         reference
35            "10.5.1.5 of IEEE Std 802.1CB-2017";
36     }
37 }
38 container organization-specific {
39     presence "true";
40     description
41        "This container allows to select Sequence encode/decode types
42        that are defined by entities outside of IEEE 802.1.";
43     reference
44        "10.5.1.5 of IEEE Std 802.1CB-2017";
45     leaf type-number {
46         type int32 {
47             range "256..max";
48         }
49         description
50            "The type number used for an encode/decode method defined by
51            an entity owning the OUI or CID for this encapsulation type.";
52         reference
53            "10.5.1.5 of IEEE Std 802.1CB-2017";
54     }
55     leaf oui-cid {
56         type string {
57             pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
58         }
59         description
```

```
1         "The Organizationally Unique Identifier (OUI) or Company ID
2         (CID) to identify the organization defining the encode/decode
3         method.";
4         reference
5         "10.5.1.5 of IEEE Std 802.1CB-2017";
6     }
7 }
8 }
9 }
10 grouping sequence-recovery-algorithm {
11     description
12     "The sequence-recovery-algorithm grouping specifies which sequence
13     recovery algorithm is to be used for this instance of the Sequence
14     recovery function.";
15     reference
16     "10.4.1.5 of IEEE Std 802.1CB-2017";
17     choice algorithm {
18         description
19         "This choice indicates the sequence recovery algorithm used for
20         this instance of the Sequence recovery function. It includes an
21         Organizationally Unique Identifier (OUI) or Company ID (CID) to
22         identify the organization defining the sequence recovery
23         algorithm.";
24         reference
25         "10.4.1.5 of IEEE Std 802.1CB-2017";
26         container vector {
27             presence "true";
28             description
29             "Vector Recovery Algorithm.";
30             reference
31             "10.4.1.5 of IEEE Std 802.1CB-2017";
32             leaf type-number {
33                 type dot1cb-frer-types:seq-rcvy-algorithm;
34                 default "vector";
35                 config false;
36                 description
37                 "The type number used for the VectorRecoveryAlgorithm.";
38                 reference
39                 "10.4.1.5 of IEEE Std 802.1CB-2017";
40             }
41             leaf oui-cid {
42                 type string {
43                     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
44                 }
45                 default "00-80-C2";
46                 config false;
47                 description
48                 "The Organizationally Unique Identifier (OUI) or Company ID
49                 (CID) to identify the organization defining the sequence
50                 recovery algorithm. For sequence recovery algorithms defined
51                 in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
52                 reference
53                 "10.4.1.5 of IEEE Std 802.1CB-2017";
54             }
55         }
56         container match {
57             presence "true";
58             description
59             "Match Recovery Algorithm.";
```

```

1      reference
2          "10.4.1.5 of IEEE Std 802.1CB-2017";
3      leaf type-number {
4          type dot1cb-frer-types:seq-rcvy-algorithm;
5          default "match";
6          config false;
7          description
8              "The type number used for the MatchRecoveryAlgorithm.";
9          reference
10             "10.4.1.5 of IEEE Std 802.1CB-2017";
11      }
12      leaf oui-cid {
13          type string {
14              pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
15          }
16          default "00-80-C2";
17          config false;
18          description
19              "The Organizationally Unique Identifier (OUI) or Company ID
20              (CID) to identify the organization defining the sequence
21              recovery algorithm. For sequence recovery algorithms defined
22              in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
23          reference
24              "10.4.1.5 of IEEE Std 802.1CB-2017";
25      }
26  }
27  container organization-specific {
28      presence "true";
29      description
30          "This container allows to select sequence recovery algorithms
31          that are defined by entities outside of IEEE 802.1.";
32      reference
33          "10.4.1.5 of IEEE Std 802.1CB-2017";
34      leaf type-number {
35          type int32 {
36              range "256..max";
37          }
38          description
39              "The type number used for a sequence recovery algorithm
40              defined by an entity owning the OUI or CID for this algorithm
41              type.";
42          reference
43              "10.4.1.5 of IEEE Std 802.1CB-2017";
44      }
45      leaf oui-cid {
46          type string {
47              pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
48          }
49          description
50              "The Organizationally Unique Identifier (OUI) or Company ID
51              (CID) to identify the organization defining the sequence
52              recovery algorithm.";
53          reference
54              "10.4.1.5 of IEEE Std 802.1CB-2017";
55      }
56  }
57  }
58  }
59  container frer {

```

description

"The managed objects that control Stream identification are described in Clause 9 of IEEE Std 802.1CB-2017. The managed objects that control FRER are described in Clause 10 of IEEE Std 802.1CB-2017 as follows:

- a) General requirements on the behavior of counters are in 10.1.
- b) The various tables of managed objects that can manage, in detail, each individual Stream, are described in five subclauses, including:
 - 1) Additions to the Stream identity table required for Autoconfiguration.
 - 2) The Sequence generation table that configures instances of the Sequence generation function;
 - 3) The Sequence recovery table that configures instances of the Individual recovery function, the Sequence recovery function, and the Latent error detection function;
 - 4) The Sequence identification table that configures instances of the Sequence encode/decode function; and
 - 5) The Stream split table that configures instances of the Stream splitting function.
- c) The managed objects that support the automatic configuration, upon receipt of a packet, of entries in the first four of the preceding tables, are described in the subclause on Autoconfiguration.
- d) The per-port, per-Stream packet counters that are kept by FRER functions for inspection by network management entities are described in 10.8, and the per-port (totaled over all Streams) counters in 10.9.

The managed objects in the subclauses under 9.1 make it possible to configure more than one encapsulation for the same stream_handle subparameter on the same port. Similarly, the managed objects in the subclauses under 10.3 and 10.4 make it possible to configure more than one Sequence encode/decode function or more than one Sequence generation function for the same stream_handle subparameter. [The same value of stream_handle can be in the frerSeqGenStreamList of more than one frerSeqGenEntry or in the frerSeqRcvyStreamList of more than one frerSeqRcvyEntry.] A system shall return an error if an attempt is made to configure conflicting requirements upon that system.";

reference

"Clause 10 of IEEE Std 802.1CB-2017";

list sequence-generation {

key "index";

description

"There is one Sequence generation table in a system, and one entry in the Sequence generation table for each Sequence generation function. Each frerSeqGenEntry lists the Streams and direction for which a single instance of the Sequence generation function is to be placed.";

reference

"10.3 of IEEE Std 802.1CB-2017";

leaf index {

type uint32;

description

"Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream handle can be present multiple times in the sequence generation table. Therefore this index leaf is being used to uniquely

```

1         identify an entry in the sequence generation table.";
2     }
3     leaf-list stream {
4         type leafref {
5             path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
6         }
7         min-elements 1;
8         description
9             "A list of stream_handle values, corresponding to the values of
10             the tsnStreamIdHandle objects in the Stream identity table, on
11             which this instance of the Sequence generation function is to
12             operate. The single instance of the Sequence generation
13             function created by this frerSeqGenEntry operates every packet
14             belonging to this Stream, regardless of the port on which it is
15             received.";
16         reference
17             "10.3.1.1 of IEEE Std 802.1CB-2017";
18     }
19     leaf direction-out-facing {
20         type dot1cb-sid-types:direction;
21         description
22             "An object indicating whether the Sequence generation function
23             is to be placed on the out-facing (True) or in-facing (False)
24             side of the port.";
25         reference
26             "10.3.1.2 of IEEE Std 802.1CB-2017";
27     }
28     leaf reset {
29         type boolean;
30         description
31             "A Boolean object indicating that the Sequence generation
32             function is to be reset by calling its corresponding
33             SequenceGenerationReset function. Writing the value True to
34             frerSeqGenReset triggers a reset; writing the value False has
35             no effect. When read, frerSeqGenReset always returns the value
36             False.";
37         reference
38             "10.3.1.3 of IEEE Std 802.1CB-2017";
39     }
40 }
41 list sequence-recovery {
42     key "index";
43     description
44         "There is one Sequence recovery table in a system, and one entry
45         in the Sequence recovery table for each Sequence recovery
46         function or Individual recovery function that can also be
47         present. The entry describes a set of managed objects for the
48         single instance of a Base recovery function and Latent error
49         detection function included in the Sequence recovery function or
50         Individual recovery function. Each frerSeqRcvyEntry lists the
51         Streams, ports, and direction for which instances of a Sequence
52         recovery function or Individual recovery function are to be
53         instantiated.";
54     reference
55         "10.4 of IEEE Std 802.1CB-2017";
56     leaf index {
57         type uint32;
58         description
59             "Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream

```

```
1      handle can be present multiple times in the sequence recovery
2      table. Therefore this index leaf is being used to uniquely
3      identify an entry in the sequence recovery table.";
4  }
5  leaf-list stream {
6      type leafref {
7          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
8      }
9      min-elements 1;
10     description
11         "A list of the stream_handle values, corresponding to the
12         values of the tsNStreamIdHandle objects in the Stream identity
13         table, to which the system is to apply the instance of the
14         Sequence recovery function or Individual recovery function.";
15     reference
16         "10.4.1.1 of IEEE Std 802.1CB-2017";
17 }
18 leaf-list port {
19     type if:interface-ref;
20     min-elements 1;
21     description
22         "The list of ports on each of which the system is to
23         instantiate the Sequence recovery function, or from which
24         received packets are to be fed to a single instance of the
25         Individual recovery function.";
26     reference
27         "10.4.1.2 of IEEE Std 802.1CB-2017";
28 }
29 leaf direction-out-facing {
30     type dot1cb-sid-types:direction;
31     description
32         "An object indicating whether the Sequence recovery function or
33         Individual recovery function is to be placed on the out-facing
34         (True) or in-facing (False) side of the port.";
35     reference
36         "10.4.1.3 of IEEE Std 802.1CB-2017";
37 }
38 leaf reset {
39     type boolean;
40     default "false";
41     description
42         "A Boolean object indicating that the Sequence recovery
43         function or Individual recovery function is to be reset by
44         calling its corresponding SequenceGenerationReset function.
45         Writing the value True to frerSeqRcvyReset triggers a reset;
46         writing the value False has no effect. When read,
47         frerSeqRcvyReset always returns the value False.";
48     reference
49         "10.4.1.4 of IEEE Std 802.1CB-2017";
50 }
51 container algorithm {
52     description
53         "This object is an enumerated value specifying which sequence
54         recovery algorithm is to be used for this instance of the
55         Sequence recovery function. The enumeration uses an OUI or CID
56         as shown in Table 10-1. The default value for
57         frerSeqRcvyAlgorithm is Vector_Alg (00-80-C2, 0).";
58     reference
59         "10.4.1.5 of IEEE Std 802.1CB-2017";
```

```
1      uses sequence-recovery-algorithm;
2  }
3  leaf history-length {
4      type sequence-history-length;
5      default "2";
6      description
7          "An integer specifying how many bits of the SequenceHistory
8          variable are to be used. The minimum and the default value is
9          2, maximum is the maximum allowed by the implementation. [Not
10         used if frerSeqRcvyAlgorithm = Match_Alg (00-80-C2, 1).]";
11     reference
12         "10.4.1.6 of IEEE Std 802.1CB-2017";
13 }
14 leaf reset-timeout {
15     type uint32;
16     units "ms";
17     description
18         "An unsigned integer specifying the timeout period in
19         milliseconds for the RECOVERY_TIMEOUT event.";
20     reference
21         "10.4.1.7 of IEEE Std 802.1CB-2017";
22 }
23 leaf invalid-sequence-value {
24     type uint32;
25     config false;
26     description
27         "A read-only unsigned integer value that cannot be encoded in a
28         packet as a value for the sequence_number subparameter, i.e.,
29         frerSeqRcvyInvalidSequenceValue is larger than or equal to
30         RecovSeqSpace.";
31     reference
32         "10.4.1.8 of IEEE Std 802.1CB-2017";
33 }
34 leaf take-no-sequence {
35     type boolean;
36     default "false";
37     description
38         "A Boolean value specifying whether packets with no
39         sequence_number subparameter are to be accepted (True) or not
40         (False). Default value False.";
41     reference
42         "10.4.1.9 of IEEE Std 802.1CB-2017";
43 }
44 leaf individual-recovery {
45     type boolean;
46     description
47         "A Boolean value specifying whether this entry describes a
48         Sequence recovery function or Individual recovery function.
49         a) True: The entry describes an Individual recovery function.
50             Packets discarded by the SequenceGenerationAlgorithm will
51             cause the variable RemainingTicks to be reset. There is no
52             Latent error detection function associated with this entry,
53             so frerSeqRcvyLatentErrorDetection cannot also be True.
54         b) False: The entry describes a Sequence recovery function.
55             Packets discarded by the SequenceGenerationAlgorithm will
56             not cause the variable RemainingTicks to be reset.";
57     reference
58         "10.4.1.10 of IEEE Std 802.1CB-2017";
59 }
```

```

1  leaf latent-error-detection {
2      type boolean;
3      description
4          "A Boolean value indicating whether an instance of the Latent
5          error detection function is to be instantiated along with the
6          Base recovery function in this Sequence recovery function or
7          Individual recovery function. frerSeqRcvyLatentErrorDetection
8          cannot be set True if frerSeqRcvyIndividualRecovery is also
9          True; an Individual recovery function does not include a Latent
10         error detection function.";
11     reference
12         "10.4.1.11 of IEEE Std 802.1CB-2017";
13 }
14 container latent-error-detection-parameters {
15     description
16         "The objects in the following subclauses are present if and
17         only if frerSeqRcvyIndividualRecovery is False.";
18     reference
19         "10.4.1.12 of IEEE Std 802.1CB-2017";
20     leaf difference {
21         type int32;
22         description
23             "An integer specifying the maximum difference between
24             frerCpsSeqRcvyDiscardedPackets, and the product of
25             frerCpsSeqRcvyPassedPackets and (frerSeqRcvyLatentErrorPaths
26             - 1) that is allowed. Any larger difference will trigger the
27             detection of a latent error by the LatentErrorTest function.";
28         reference
29             "10.4.1.12.1 of IEEE Std 802.1CB-2017";
30     }
31     leaf period {
32         type uint32;
33         units "ms";
34         default "2000";
35         description
36             "The integer number of milliseconds that are to elapse
37             between instances of running the LatentErrorTest function. An
38             implementation can have a minimum value for
39             frerSeqRcvyLatentErrorPeriod, below which it cannot be set,
40             but this minimum shall be no larger than 1000 ms (1 s).
41             Default value 2000 (2 s).";
42         reference
43             "10.4.1.12.2 of IEEE Std 802.1CB-2017";
44     }
45     leaf paths {
46         type uint16;
47         description
48             "The integer number of paths over which FRER is operating for
49             this instance of the Base recovery function and Latent error
50             detection function.";
51         reference
52             "10.4.1.12.3 of IEEE Std 802.1CB-2017";
53     }
54     leaf reset-period {
55         type uint32;
56         units "ms";
57         default "30000";
58         description
59             "The integer number of milliseconds that are to elapse

```

```
1         between instances of running the LatentErrorReset function.
2         An implementation can have a minimum value for
3         LatentErrorReset, below which it cannot be set, but this
4         minimum shall be no larger than 1000 ms (1 s). Default value
5         30000 (30 s).";
6         reference
7         "10.4.1.12.4 of IEEE Std 802.1CB-2017";
8     }
9 }
10 }
11 list sequence-identification {
12     key "port direction-out-facing";
13     description
14         "There is one Sequence identification table per system, and one
15         entry in the Sequence identification table for each port and
16         direction for which an instance of the Sequence encode/decode
17         function is to be created. Each entry in the Sequence
18         identification table specifies a port and direction on which an
19         instance of the Sequence encode/decode function is to be
20         instantiated for a list of Streams.";
21     reference
22         "10.5 of IEEE Std 802.1CB-2017";
23     leaf-list stream {
24         type leafref {
25             path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
26         }
27         min-elements 1;
28         description
29             "A list of stream_handles, corresponding to the values of the
30             tsnStreamIdHandle objects in the Stream identity table, for
31             which the system is to use the same encapsulation for the
32             Sequence encode/decode function.";
33         reference
34             "10.5.1.1 of IEEE Std 802.1CB-2017";
35     }
36     leaf port {
37         type if:interface-ref;
38         description
39             "The port on which the system is to place an instance of the
40             Sequence encode/decode function.";
41         reference
42             "10.5.1.2 of IEEE Std 802.1CB-2017";
43     }
44     leaf direction-out-facing {
45         type dot1cb-sid-types:direction;
46         description
47             "An object indicating whether the Sequence encode/decode
48             function is to be placed on the out-facing (True) or in-facing
49             (False) side of the port.";
50         reference
51             "10.5.1.3 of IEEE Std 802.1CB-2017";
52     }
53     leaf active {
54         type boolean;
55         description
56             "A Boolean value specifying whether this frerSeqEncEntry is
57             passive (False), and therefore is used only to decode (extract
58             information from) input packets passing up the protocol stack,
59             or active (True), and therefore is used both for recognizing
```

```

1      input packets and for encoding output packets being passed down
2      the protocol stack.";
3      reference
4      "10.5.1.4 of IEEE Std 802.1CB-2017";
5  }
6  container encapsulation {
7      description
8      "An enumerated value indicating the type of encapsulation used
9      for this instance of the Sequence encode/decode function. The
10     type includes an OUI or CID.";
11     reference
12     "10.5.1.5 of IEEE Std 802.1CB-2017";
13     uses sequence-encode-decode;
14 }
15 leaf path-id-lan-id {
16     type lan-path-id;
17     description
18     "A 4-bit integer value to be placed in the PathId field of an
19     HSR sequence tag or the LanId field of a PRP sequence trailer
20     added to an output packet. This managed object is used only if:
21     a) The HSR sequence tag or the PRP sequence trailer is
22     selected by the frerSeqEncEncapsType object; and
23     b) frerSeqEncActive is False (passive)";
24     reference
25     "10.5.1.6 of IEEE Std 802.1CB-2017";
26 }
27 }
28 list stream-split {
29     key "port-direction-out-facingindex";
30     description
31     "There is one Stream split table per system, with one
32     frerSplitEntry per Stream splitting function per set of
33     stream_handle values. Each entry in the Stream split table
34     specifies a port and direction on which an instance of the Stream
35     splitting function is to be instantiated, and the list of
36     stream_handles specifying its operation.";
37     reference
38     "10.6 of IEEE Std 802.1CB-2017";
39     leaf index {
40         type uint32;
41         description
42         "Entry in the Stream split table.";
43         reference
44         "10.6 of IEEE Std 802.1CB-2017";
45     }
46     leaf port {
47         type if:interface-ref;
48         description
49         "The port on which the system is to place an instance of the
50         Stream splitting function performing the stream_handle
51         translations specified by frerSplitInputIdList and
52         frerSplitOutputIdList is to be placed.";
53         reference
54         "10.6.1.1 of IEEE Std 802.1CB-2017";
55     }
56     leaf direction-out-facing {
57         type dot1cb-sid-types:direction;
58         description
59         "An object indicating whether the instance of the Stream

```

```
1      splitting function performing the stream_handle translations
2      specified by frerSplitInputIdList and frerSplitOutputIdList is
3      to be placed on the out-facing (True) or in-facing (False) side
4      of the port.";
5      reference
6      "10.6.1.2 of IEEE Std 802.1CB-2017";
7  }
8  leaf-list input-id {
9      type leafref {
10         path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
11     }
12     min-elements 1;
13     description
14         "A list of stream_handles (tsnStreamIdHandle values) that are
15         to be split.";
16     reference
17         "10.6.1.3 of IEEE Std 802.1CB-2017";
18 }
19 leaf-list output-id {
20     type leafref {
21         path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
22     }
23     min-elements 1;
24     description
25         "A list of stream_handles (tsnStreamIdHandle values) into which
26         the input packet is to be split, one copy per item in the
27         frerSplitOutputIdList.";
28     reference
29         "10.6.1.4 of IEEE Std 802.1CB-2017";
30 }
31 }
32 container autoconfiguration {
33     if-feature "auto-configuration";
34     description
35         "Container for autoconfiguration managed objects.";
36     reference
37         "10.7 of IEEE Std 802.1CB-2017";
38     list sequence {
39         key "index";
40         description
41             "There is one Sequence autoconfiguration table per system. It
42             contains any number of table entries. No two (or more) entries
43             in the Sequence autoconfiguration table can have the same
44             values for frerAutSeqSeqEncaps, frerAutSeqTagged, and
45             frerAutSeqVlan on any given port. Each frerAutSeqEntry object
46             relates to a single class of Streams, and specifies how entries
47             are created (and destroyed) in the Stream identity table, the
48             Sequence recovery table, and the Sequence identification table.";
49         reference
50             "10.7.1 of IEEE Std 802.1CB-2017";
51         leaf index {
52             type uint32;
53             description
54                 "Entry in the sequence list referencing to a single class of
55                 Streams.";
56         }
57         container sequence-encapsulation {
58             description
59                 "An enumerated value, specifying which Sequence encode/decode
```

```
1         function, and therefore, which type sequence_number encoding,
2         is to be recognized for the purposes of Autoconfiguration.";
3     reference
4         "10.7.1.1.1 of IEEE Std 802.1CB-2017";
5     uses sequence-encode-decode;
6 }
7 leaf-list receive-port {
8     type if:interface-ref;
9     min-elements 1;
10    description
11        "The list of ports to which this frerAutSeqEntry applies, and
12        on which Stream identification functions, Sequence
13        encode/decode functions, and Individual recovery functions
14        are to be autocreated.";
15    reference
16        "10.7.1.1.2 of IEEE Std 802.1CB-2017";
17 }
18 leaf tagged {
19     type enumeration {
20         enum tagged {
21             value 1;
22             description
23                 "A frame must have a VLAN tag to be matched.";
24         }
25         enum priority {
26             value 2;
27             description
28                 "A frame must be untagged, or have a VLAN tag with a VLAN
29                 ID = 0 to be matched.";
30         }
31         enum all {
32             value 3;
33             description
34                 "A frame is matched whether tagged or not.";
35         }
36     }
37    description
38        "An enumerated value indicating whether packets to be matched
39        by this frerAutSeqEntry are permitted to have a VLAN tag.";
40    reference
41        "10.7.1.1.3 of IEEE Std 802.1CB-2017";
42 }
43 leaf-list vlan {
44     type dot1qtotypes:vlanid;
45    description
46        "A list of vlan_identifiers for the packet to match. A null
47        list matches all vlan_identifiers.";
48    reference
49        "10.7.1.1.4 of IEEE Std 802.1CB-2017";
50 }
51 leaf-list recovery-port {
52     type if:interface-ref;
53     min-elements 1;
54    description
55        "The list of ports on which Sequence recovery functions are
56        to be autocreated by this frerAutSeqEntry.";
57    reference
58        "10.7.1.1.5 of IEEE Std 802.1CB-2017";
59 }
```

```
1  leaf destruction-interval {
2      type uint64;
3      units "ms";
4      default "86400000";
5      description
6          "An integer number of milliseconds after which an idle set of
7          functions created by this frerAutSeqEntry can be destroyed. A
8          value of 0 indicates that idle autoconfigured functions are
9          not to be destroyed. Default value is 86 400 000 decimal (one
10         day).";
11     reference
12         "10.7.1.1.6 of IEEE Std 802.1CB-2017";
13 }
14 leaf reset-interval {
15     type uint64;
16     units "ms";
17     description
18         "The value used to fill frerSeqRcvyResetMSec when
19         autoconfiguring entries in the Sequence recovery table.";
20     reference
21         "10.7.1.1.7 of IEEE Std 802.1CB-2017";
22 }
23 container algorithm {
24     description
25         "The value used to fill frerSeqRcvyAlgorithm when
26         autoconfiguring entries in the Sequence recovery table.";
27     reference
28         "10.7.1.1.8 of IEEE Std 802.1CB-2017";
29     uses sequence-recovery-algorithm;
30 }
31 leaf history-length {
32     type sequence-history-length;
33     default "2";
34     description
35         "The value used to fill frerSeqRcvyHistoryLength when
36         autoconfiguring entries in the Sequence recovery table.";
37     reference
38         "10.7.1.1.9 of IEEE Std 802.1CB-2017";
39 }
40 leaf create-individual {
41     type boolean;
42     description
43         "A Boolean value. If True, the receipt of a packet that
44         triggers the autoconfiguration of a new tsNStreamIdEntry also
45         triggers the instantiation of a frerSeqRcvyEntry for an
46         Individual recovery function.";
47     reference
48         "10.7.1.1.10 of IEEE Std 802.1CB-2017";
49 }
50 leaf create-recovery {
51     type boolean;
52     description
53         "A Boolean value. If True, the receipt of a packet that
54         triggers the autoconfiguration of a new tsNStreamIdEntry can
55         also trigger the instantiation of a frerSeqRcvyEntry for a
56         Sequence recovery function.";
57     reference
58         "10.7.1.1.11 of IEEE Std 802.1CB-2017";
59 }
```

```

1  leaf latent-error-detection {
2      type boolean;
3      description
4          "A Boolean value. If True, the autoconfiguration of a new
5          Sequence recovery function also creates an associated Latent
6          Error Detection function.";
7      reference
8          "10.7.1.1.12 of IEEE Std 802.1CB-2017";
9  }
10 leaf latent-error-difference {
11     type int32;
12     description
13         "The value used to fill frerSeqRcvyLatentErrorDifference when
14         autoconfiguring entries in the Sequence recovery table.";
15     reference
16         "10.7.1.1.13 of IEEE Std 802.1CB-2017";
17 }
18 leaf latent-error-period {
19     type uint32;
20     units "ms";
21     default "2000";
22     description
23         "The value used to fill frerSeqRcvyLatentErrorPeriod when
24         autoconfiguring entries in the Sequence recovery table.";
25     reference
26         "10.7.1.1.14 of IEEE Std 802.1CB-2017";
27 }
28 leaf latent-error-reset-period {
29     type uint32;
30     units "ms";
31     default "30000";
32     description
33         "The value used to fill frerSeqRcvyLatentResetPeriod when
34         autoconfiguring entries in the Sequence recovery table.";
35     reference
36         "10.7.1.1.15 of IEEE Std 802.1CB-2017";
37 }
38 }
39 list output {
40     key "index";
41     description
42         "There is one Output autoconfiguration table per system. It
43         contains any number of frerAutOutEntry objects, each relating
44         to a single class of Streams specifying how active entries are
45         created in the Sequence identification table. No two (or more)
46         entries in the Output autoconfiguration table can include the
47         same port in their frerAutSeqReceivePortList objects.";
48     reference
49         "10.7.2 of IEEE Std 802.1CB-2017";
50     leaf index {
51         type uint32;
52         description
53             "Entry in the output list referencing to a single class of
54             Streams.";
55     }
56     leaf-list port {
57         type if:interface-ref;
58         min-elements 1;
59         description

```

```

1         "The list of ports to which this frerAutOutEntry applies, and
2         on which active Sequence encode/decode functions are to be
3         autocreated.";
4         reference
5         "10.7.2.1.1 of IEEE Std 802.1CB-2017";
6     }
7     container encapsulation {
8         description
9         "An enumerated value, specifying which Sequence encode/decode
10        function, and therefore, which type sequence_number encoding,
11        is to be used for autoconfigured Streams on the ports in
12        frerAutSeqReceivePortList.";
13        reference
14        "10.7.2.1.2 of IEEE Std 802.1CB-2017";
15        uses sequence-encode-decode;
16    }
17    leaf lan-path-id {
18        type lan-path-id;
19        description
20        "An integer specifying a path or LAN. If and only if
21        frerAutOutEncaps specifies HSR or PRP frerAutOutLanPathId
22        specifies the LanId or PathId value to be inserted into the
23        HSR sequence tag or PRP sequence trailer of autoconfigured
24        packets transmitted on the ports in
25        frerAutSeqReceivePortList.";
26        reference
27        "10.7.2.1.3 of IEEE Std 802.1CB-2017";
28    }
29 }
30 }
31 }
32 augment "/dot1cb-sid:stream-identity" {
33     if-feature "auto-configuration";
34     description
35     "Two managed objects augment each tsNStreamIdEntry in the Stream
36     identity table when Managed objects for autoconfiguration is
37     implemented.";
38     reference
39     "10.2 of IEEE Std 802.1CB-2017";
40     leaf auto-configured {
41         type boolean;
42         config false;
43         description
44         "A read-only Boolean value, supplied by the system, specifying
45         whether this entry was created explicitly (False) or via the
46         Sequence autoconfiguration table.";
47         reference
48         "10.2.1 of IEEE Std 802.1CB-2017";
49     }
50     leaf lan-path-id {
51         type lan-path-id;
52         description
53         "An integer specifying a path or LAN. If and only if a packet
54         matches an entry in the Sequence identification table that
55         specifies HSR or PRP in its frerSeqEncEncapsType object,
56         tsNStreamIdLanPathId specifies the LanId or PathId value that
57         must be matched for this tsNStreamIdEntry to apply. A value of -1
58         indicates that the LanId or PathId are to be ignored.";
59         reference

```

```
1      "10.2.2 of IEEE Std 802.1CB-2017";
2    }
3  }
4  augment "/if:interfaces/if:interface/if:statistics" {
5    description
6      "The following counters are the counters for frame replication and
7       elimination for reliability. All counters are unsigned integers. If
8       used on links faster than 650 000 000 bits per second, they shall
9       be 64 bits in length to ensure against excessively short wrap
10      times.
11
12      A Stream identification component shall implement the first two
13      counters provided in the stream-identification YANG module per-port
14      counters, input-packets and output-packets; the remainder of the
15      counters in the frer YANG module module are optional for such a
16      system.";
17    reference
18      "10.8 of IEEE Std 802.1CB-2017
19       10.9 of IEEE Std 802.1CB-2017";
20    container frer {
21      description
22        "This container contains the per-port as well as the
23         per-port-per-stream counters for frame replication and
24         elimination for reliability.";
25      reference
26        "10.8 of IEEE Std 802.1CB-2017
27         10.9 of IEEE Std 802.1CB-2017";
28      container per-port-counters {
29        config false;
30        description
31          "Contains the per-port counters for frame replication and
32           elimination for reliability. The following counters are
33           instantiated for each port on which any of the Stream
34           identification function, Sequencing function, or Sequence
35           encode/decode function is configured. The counters are indexed
36           by port number.";
37        leaf rx-passed-pkts {
38          type uint64;
39          config false;
40          description
41            "The frerCpSeqRcvyPassedPackets counter is incremented once
42             for each packet passed up the stack by the
43             VectorRecoveryAlgorithm or MatchRecoveryAlgorithm. Its value
44             equals the sum (modulo the size of the counters) of all of
45             the frerCpsSeqRcvyPassedPackets counters on this same port.";
46          reference
47            "10.9.1 of IEEE Std 802.1CB-2017";
48        }
49        leaf rx-discarded-pkts {
50          type uint64;
51          config false;
52          description
53            "The frerCpSeqRcvyDiscardPackets counter is incremented once
54             for each packet discarded due to a duplicate sequence number
55             or for being a rogue packet by any VectorRecoveryAlgorithm or
56             MatchRecoveryAlgorithm on this port. Its value equals the sum
57             (modulo the size of the counters) of all of the
58             frerCpsSeqRcvyRoguePackets and frerCpsSeqRcvyDiscardedPackets
59             counters on this same port.";
```

```
1      reference
2        "10.9.2 of IEEE Std 802.1CB-2017";
3    }
4    leaf encode-errored-pkts {
5      type uint64;
6      config false;
7      description
8        "The frerCpSeqEncErroredPackets counter is incremented once
9        each time the Sequence encode/decode function receives a
10       packet that it is unable to decode successfully. Its value
11       equals the sum (modulo the size of the counters) of all of
12       the frerCpsSeqEncErroredPackets counters on this same port.";
13      reference
14        "10.9.2 of IEEE Std 802.1CB-2017";
15    }
16  }
17  list per-port-per-stream-counters {
18    key "direction-out-facing handle";
19    config false;
20    description
21      "Contains the per-port-per-stream counters for frame
22      replication and elimination for reliability. The following
23      counters are instantiated for each port on which any of the
24      Stream identification function, Sequencing function, or
25      Sequence encode/decode function is configured. The counters are
26      indexed by port number, facing (in-facing or out-facing), and
27      stream_handle value.";
28    reference
29      "10.8 of IEEE Std 802.1CB-2017";
30    leaf direction-out-facing {
31      type dot1cb-sid-types:direction;
32      description
33        "An object indicating whether the counters apply to
34        out-facing (True) or in-facing (False).";
35    }
36    leaf handle {
37      type leafref {
38        path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
39      }
40      description
41        "The according tsNStreamIdHandle for the counters.";
42    }
43    leaf generation-reset {
44      type uint64;
45      config false;
46      description
47        "The frerCpsSeqGenResets counter is incremented each time the
48        SequenceGenerationReset function is called.";
49      reference
50        "10.8.2 of IEEE Std 802.1CB-2017";
51    }
52    leaf rx-out-of-order-pkts {
53      type uint64;
54      config false;
55      description
56        "The frerCpsSeqRcvyOutOfOrderPackets counter is incremented
57        once for each packet accepted out-of-order by the
58        VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.
59        Out-of-order means that the packet's sequence number is not
```

```
1         one more than the previous packet received.";
2     reference
3         "10.8.3 of IEEE Std 802.1CB-2017";
4 }
5 leaf rx-rogue-pkts {
6     type uint64;
7     config false;
8     description
9         "The frerCpsSeqRcvyRoguePackets counter is incremented once
10        for each packet discarded by the VectorRecoveryAlgorithm
11        because its sequence_number subparameter is more than
12        frerSeqRcvyHistoryLength from RecovSeqNum.";
13    reference
14        "10.8.4 of IEEE Std 802.1CB-2017";
15 }
16 leaf rx-passed-pkts {
17     type uint64;
18     config false;
19     description
20        "The frerCpsSeqRcvyPassedPackets counter is incremented once
21        for each packet passed up the stack by the
22        VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.";
23    reference
24        "10.8.5 of IEEE Std 802.1CB-2017";
25 }
26 leaf rx-discarded-pkts {
27     type uint64;
28     config false;
29     description
30        "The frerCpsSeqRcvyDiscardedPackets counter is incremented
31        once for each packet discarded due to a duplicate sequence
32        number by the VectorRecoveryAlgorithm or
33        MatchRecoveryAlgorithm.";
34    reference
35        "10.8.6 of IEEE Std 802.1CB-2017";
36 }
37 leaf rx-lost-pkts {
38     type uint64;
39     config false;
40     description
41        "The frerCpsSeqRcvyLostPackets counter is incremented once
42        for each packet lost by the VectorRecoveryAlgorithm. A packet
43        is counted as lost if its sequence number is not received on
44        any ingress port.
45
46        NOTE—If per-source sequence numbering is used,
47        frerCpsSeqRcvyLostPackets can count, as lost, packets that
48        were sent to another destination, but not lost.";
49    reference
50        "10.8.7 of IEEE Std 802.1CB-2017";
51 }
52 leaf rx-tagless-pkts {
53     type uint64;
54     config false;
55     description
56        "The frerCpsSeqRcvyTaglessPackets counter is incremented once
57        for each packet received by the VectorRecoveryAlgorithm that
58        has no sequence_number subparameter.";
59    reference
```

```

1      "10.8.8 of IEEE Std 802.1CB-2017";
2  }
3  leaf rx-resets {
4      type uint64;
5      config false;
6      description
7          "The frerCpsSeqRcvyResets counter is incremented once each
8              time the SequenceRecoveryReset function is called.";
9      reference
10         "10.8.9 of IEEE Std 802.1CB-2017";
11 }
12 leaf rx-latent-error-resets {
13     type uint64;
14     config false;
15     description
16         "The frerCpsSeqRcvyLatentErrorResets counter is incremented
17             once each time the LatentErrorReset function is called.";
18     reference
19         "10.8.10 of IEEE Std 802.1CB-2017";
20 }
21 leaf encode-errored-pkts {
22     type uint64;
23     config false;
24     description
25         "The frerCpsSeqEncErroredPackets counter is incremented once
26             each time the Sequence encode/decode function receives a
27             packet that it is unable to decode successfully.";
28     reference
29         "10.8.11 of IEEE Std 802.1CB-2017";
30 }
31 }
32 }
33 }
34 }
35
36
37

```